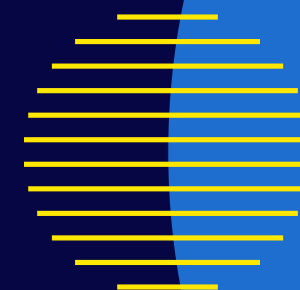


INTRODUCTION TO WEB TECHNOLOGIES

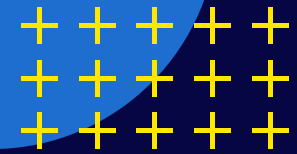
PPT ON UNIT I FOR WEB APPLICATION
DEVELOPMENT

PES MODERN COLLEGE OF ENGINEERING, PUNE

Presented by Ashish Singh and
Rushabh Bhalgat under the guidance
of Mrs. Supriya Jagtap



GETTING STARTED WITH HTML



- HTML stands for HyperText Markup Language and is used to structure web documents
- It serves as the skeleton of all web pages by organizing content like text, images, and multimedia
- Web browsers interpret HTML to render visually meaningful web pages to users
- Known for its simple, tag-based syntax that is both human-readable and easy to learn

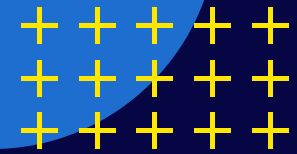
HTML



WHY HTML?

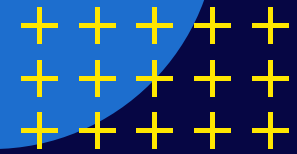
- HTML is the foundational language for structuring web pages and web applications
- It enables the integration of multimedia content and interaction with other web technologies
- Universally supported across all modern browsers and devices, ensuring maximum compatibility
- Works in harmony with CSS for styling and JavaScript for dynamic functionality

TAGS AND ELEMENTS



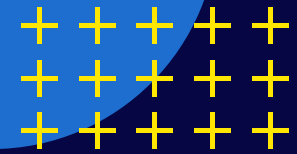
- HTML uses tags enclosed in angle brackets (e.g., `<tag>`) to define elements on a web page
- Elements usually include a start tag, content, and an end tag (e.g., `<tag>This is a paragraph</tag>`)
- Some tags are self-closing like `<br />` or `<img alt="image" />`, which don't require an end tag
- Common tags are used for structuring content such as headings, links, images, and tables

ATTRIBUTES AND PROPERTIES



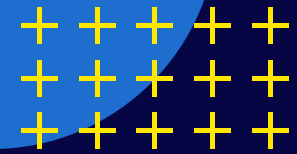
- Attributes add additional information to HTML elements and follow a name="value" format
- Examples include src, alt, href, id, class, and style, used to define sources, links, and appearance
- The style attribute can apply inline CSS directly to elements for quick customization
- Proper use of attributes enhances accessibility, SEO, and interactivity

HEADINGS AND LISTS



- HTML headings range from (most important) to (least important) and define content hierarchy
- Headings are crucial for organizing content, SEO optimization, and screen readers
- Lists include ordered lists () for sequential data and unordered lists () for non-sequential
- Each list item is enclosed in a tag; nested lists are possible for creating sub-levels

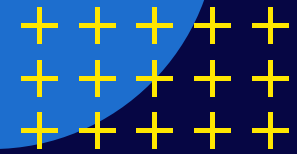
LINKS, TABLES, AND IMAGES



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HTML FORMS AND MEDIA



- Forms gather user input using `<input>`, `<select>`, and `<form>`
- Input types include: text, email, password, checkbox, radio, file, etc.
- Audio and video are embedded with `<audio>` and `<video>` tags with controls attribute
- Support multiple file types for browser compatibility and responsive playback

SEMANTIC HTML5 ELEMENTS

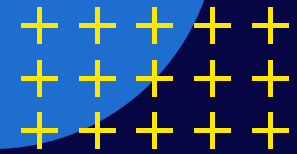
- Semantic tags like `<h1>`, `<h2>`, `<h3>`, `<h4>`, `<h5>`, `<h6>` define roles
- Improve structure, readability, accessibility, and SEO
- Help screen readers and developers understand page layout
- Promote standardized, maintainable, and future-proof code

WHY CSS?

- CSS styles HTML content visually by defining design rules
- Enables layout control, colors, fonts, animations, and transitions
- Allows consistent and professional look across multiple pages
- Encourages separation of content (HTML) and design (CSS)



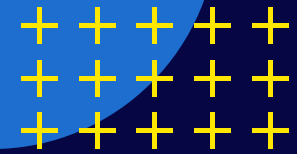
TYPES OF CSS



- Inline CSS: Applied directly to elements with style="..."
- Internal CSS: Included inside tags within HTML's
- External CSS: Linked via to separate .css file for reusable global styles
- External stylesheets are best for clean code and scalability

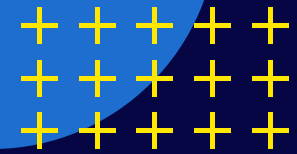


CSS PROPERTIES AND CLASSES



- Properties control layout, fonts, colors, backgrounds, margins, padding, etc.
- Classes (.) are reusable across multiple elements
- IDs (#) uniquely identify and style individual elements
- Combine selectors for targeted and efficient styling

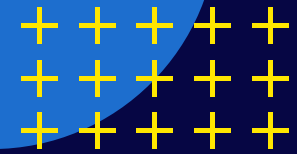
NESTED CSS (CHILD CLASS)



- Nested rules apply styles within a parent-child relationship: `.parent .child { }`
- Helps manage styles specific to sections or components
- Prevents conflicts and supports modular design in complex pages
- Improves maintainability and readability of large stylesheets



BOOTSTRAP AND W3C



- Why Bootstrap: A powerful CSS framework for responsive and mobile-first design
- Using Bootstrap: Link CDN or download files; use built-in classes for rapid design
- Grid System: Layout using rows and 12-column grid system for responsive design
- Components: Pre-designed elements like buttons, navbars, modals, alerts, tables, etc.
- W3C: World Wide Web Consortium, develops web standards and protocols
- W3C Role: Ensures web technologies are compatible, accessible, and standardized
- Promotes open web platform for universal web development



THANK YOU

